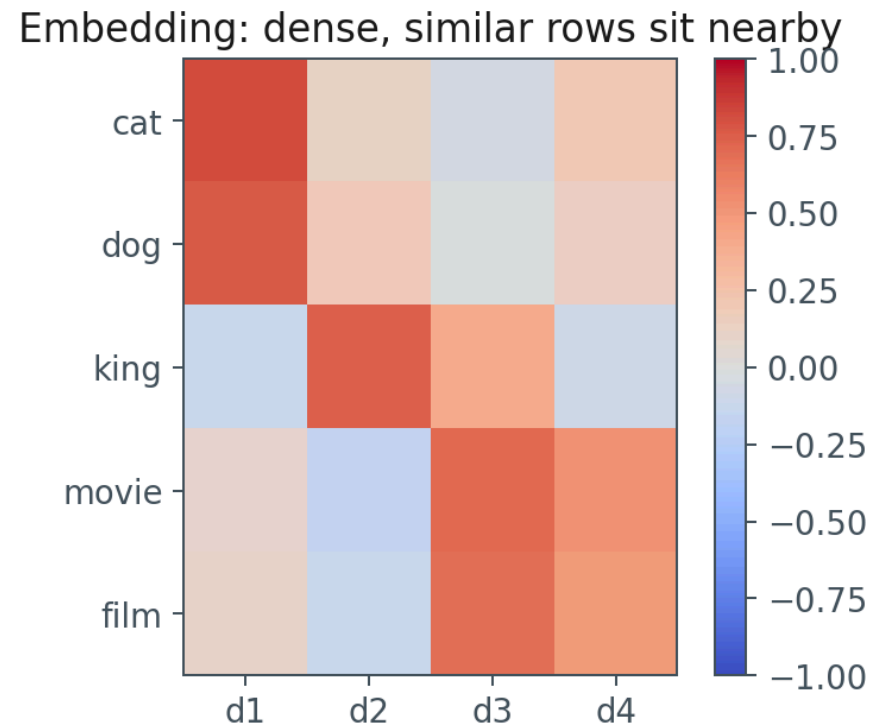
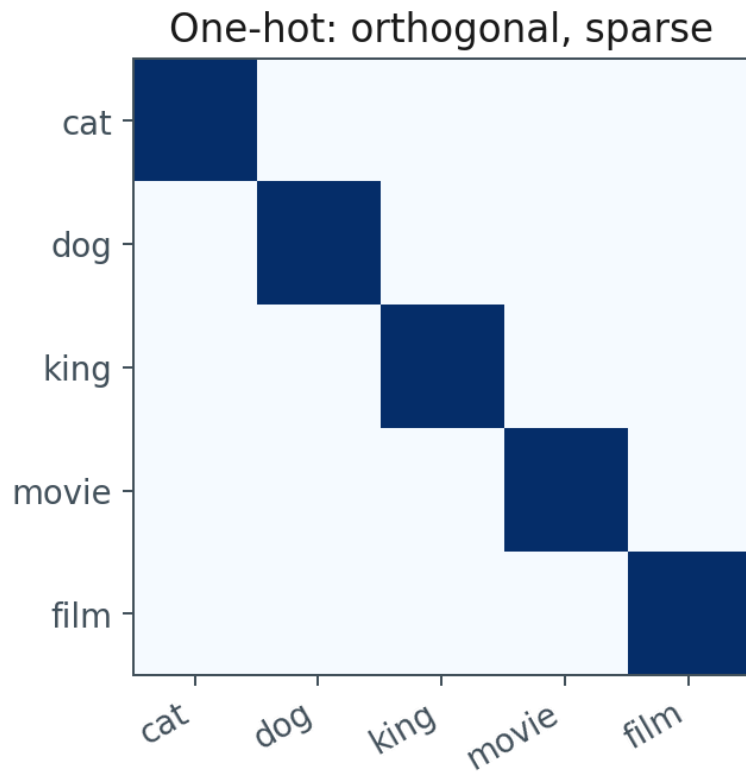


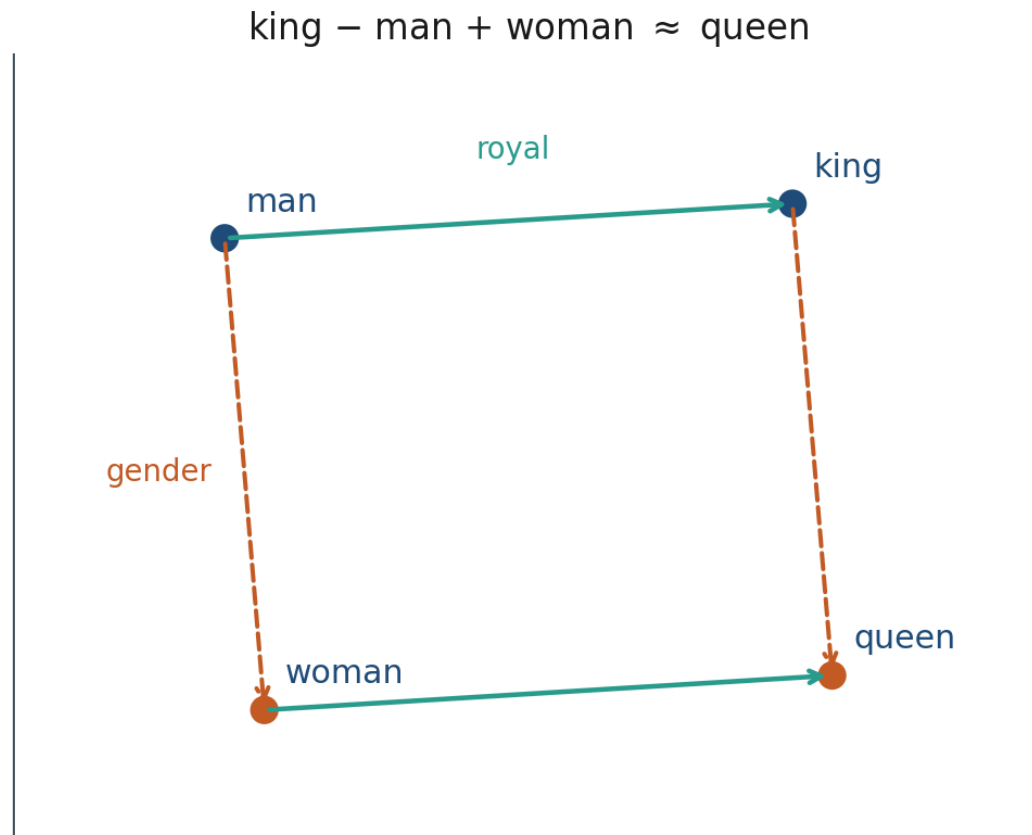
One-hot is a bad geometry

- film and movie are orthogonal. New words have no coordinate.
- Distributed = short, dense. Similar usage → similar vectors.
- An embedding is the map from a token id into that space.



Direction can mean something

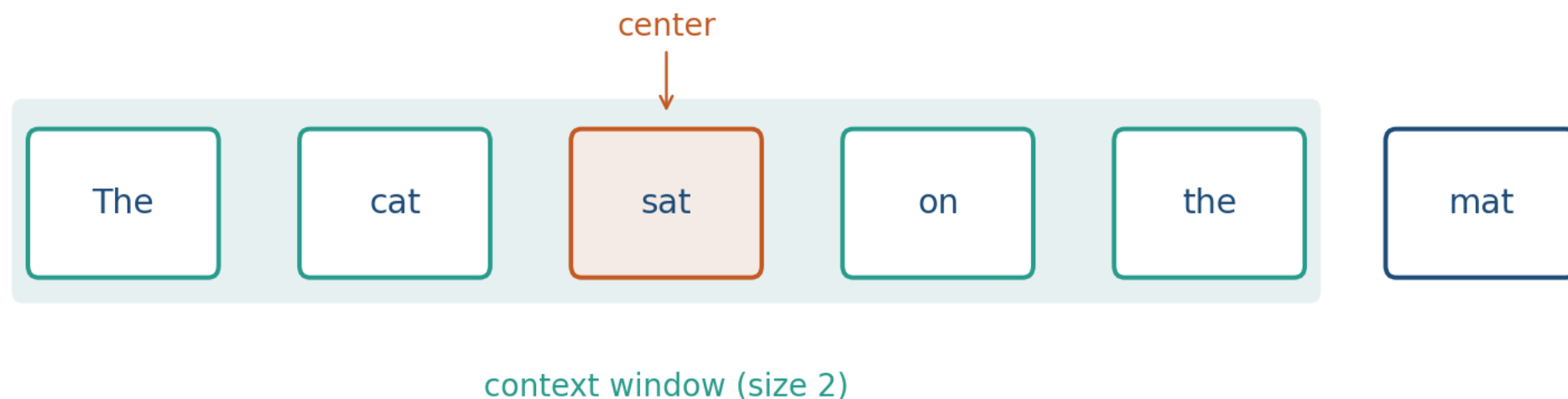
- $\text{king} - \text{man} + \text{woman} \approx \text{queen}$.
- Cosine, not Euclidean length (length tracks frequency).
- Nearest neighbors are a debug tool.



Skip-gram learns the map

- Fake task: from the center, predict each neighbor.
- Keep the hidden weights. Throw away the softmax.
- A transformer keeps transforming those vectors instead.

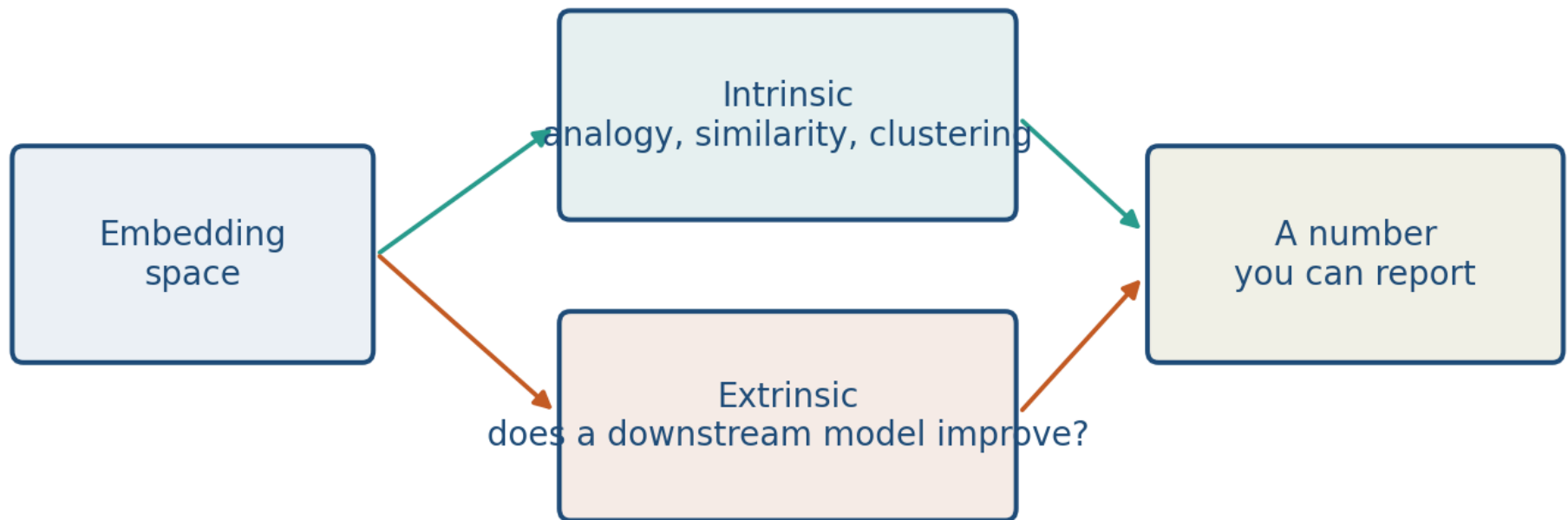
Skip-gram: predict each neighbor from the center word



Intrinsic vs. extrinsic

- Intrinsic: analogies, similarity, clustering of the space.
- Extrinsic: does a downstream model improve?
- They need not agree. Projects report extrinsic.

Two ways to score embeddings. They need not agree.



Bias is also geometry

- The same offsets can encode stereotypes.
- Probe occupations along a she/he direction.
- Lab 1 does this on a constructed 2-D space first.

